Swagat Sahu

Roll No.: 240103134
Mechatronics and Automation Engineering
Bachelors in Technology
Indian Institute Of Information Technology, Bhagalpur

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Portfolio
GitHub Profile
LinkedIn Profile

Education

•Indian Institute of Information Technology Bhagalpur *2028*
Bachelor of Technology Mechatronics and Automation Engineering CGPA/Percentage: 8

•Indian Institute of Technology Ropar *2025*
Minor in AI and ML CGPA/Percentage: 8

•St. De Paul School, 12th *2024*
Indian School Certificate (ISC) CGPA/Percentage: 91

Summary

Backend-focused developer experienced in building scalable real-time and distributed systems using AWS, Docker, and WebSockets. Skilled in designing low-latency, stateless backend architectures and REST APIs for cloud-native applications. Interested in system design, AI, and high-performance computing systems.

Technical Skills and Interests

Languages: JavaScript, Python, C++, Java, Go
Frameworks: React, Next.js, Node.js, FastAPI, Express
Tools: Docker, Git, Redis, Grafana
ML/AI: PyTorch, OpenCV, scikit-learn, LangChain
Cloud: AWS (ECS, ECR, Fargate, EC2), Google Cloud Platform (GCP)
Soft Skills: Communication, Teamwork
Areas of Interest: Distributed Systems, AI, Image Processing, System Design, Data Structures and Algorithms, Robotics, IoT

Projects

•Live Code Editor - Containerized and Deployed on AWS *03/2026*
Real-time collaborative code editor enabling multi-user editing
- Built a real-time collaborative editor using React and Yjs with concurrent multi-user editing
- Deployed scalable containerized backend using AWS ECS, Fargate, ECR and EC2 with custom VPC and load balancer
- Implemented WebSocket-based low-latency synchronization in a distributed auto scalable system with monitoring
- Live: http://new-alb-docker-1800000494.ap-south-1.elb.amazonaws.com/

•Anonymous Real-Time Chat Platform *11/2025*
Anonymous chat system with real-time communication
- Developed real-time chat platform using Next.js, WebSockets based, supporting concurrent users
- Integrated Redis for session management and real-time message broadcasting
- Designed scalable low-latency backend ensuring consistent performance under concurrent load
- Live: https://rec-red.vercel.app/

•Text-to-SQL Transformer – Custom 11.2M Parameter SLM *04/2026*
Custom-built Sequence-to-Sequence model for Natural Language to SQL translation
- Developed a 11.2M parameter Encoder-Decoder Transformer from scratch using PyTorch with a 4-layer depth.
- Implemented 8-head Multi-Head Attention ($d_{model} = 256$) and a custom BPE tokenizer for 5k vocabulary mapping.
- Optimized local CUDA training on 30k pairs using Teacher Forcing and causal masking to achieve 1.39 loss.
- GitHub: https://github.com/crocodilelurker/sql_transformer | Live: https://sqltranslate.streamlit.app/

Certification

•Postman API Fundamentals Student Expert *06/2025*

Positions of Responsibility

•Student Coordinator Training and Placement Cell, IIIT Bhagalpur *2025–Present*
- On boarded 10+ companies for placement drives, expanding recruiter outreach and increasing opportunities for students

Achievements

•Achievement Reliance Undergraduate Scholarship, Reliance Group *2025*

•Achievement Qualified for Smart India Hackathon (National Level) *2025*